STOR 320: Methods for Data Science

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Introduction

The COVID-19 pandemic severely affected the jobs, schooling, and overall lives of many people. Unemployment in the US rose from 3.7% in 2019 pre-COVID to 8.1% in 2020. Schools shut down and switched to online for nearly a year and a half, resulting in decreased literacy rates among students. Furthermore, there has been over one hundred million COVID cases, and over a million deaths from the illness. Since the pandemic has affected different groups of people in various ways, we wanted to analyze the effects of COVID-19 on different income and household groups in Chicago, Illnois for the years 2020-2021 using datasets from the City of Chicago and Kaggle. The owners of these data may benefit from our analyses by piecing together information and taking into account the different factors as to why certain trends, if any, exist.

The first question to explore is "Do ZIP codes with lower median household incomes have higher COVID-19 death/mortality rates?". This topic is interesting because the answer will allow us and the general public to search for deeper insight about why there would be a relationship, if any, between these two variables. We hypothesized that households income and the death rates from COVID-19 are negatively correlated because those with relatively more financial stability may have a wider selection of resources to access to the healthcare needed to treat positive cases.

The second question is "Do ZIP codes with lower median household incomes have higher cumulative COVID-19 case rates?". Although similar to the first question, this question attempts to promote inquisitivity. It may be reasonable to hypothesize that these variables are also negatively correlated, but there are also other factors that could influence the results. Multiple income brackets could have equal opportunity to limit their interactions with people that tested positive for COVID-19. Additionally, ZIP codes with lower household incomes may not have had the resources to obtain COVID-19 testing, which could lead to a relatively "lower" cumulative case rate due to a relatively lower testing rate.

Therefore, we found it necessary to explore the depths of what answers these questions can bring and how those in Chicago, as well as the general public, can use this knowledge to form health-related decisions in the future.

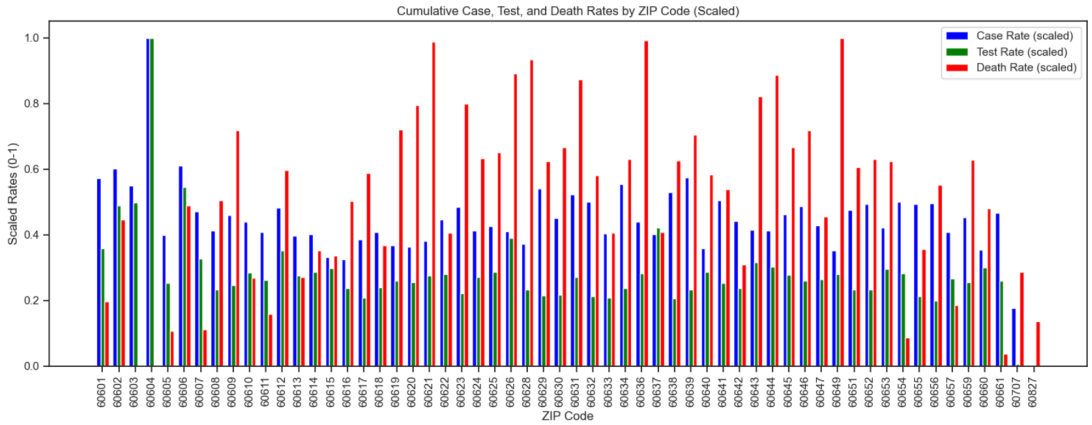
Data

The first dataset used in this report is "COVID-19 Cases, Tests, and Deaths by ZIP Code - Historical" from the City of Chicago. This data combines data the Illinois National Electronic Disease Surveillance System, Cook County Medical Examiner's Office, Illinois Vital Records, and the American Community Survey (US Census). The second dataset is "US Household Income by Zip Code 2021-2011" collected by CLAYGENDRON in Kaggle. This dataset combined the 2011 and 2021 US Census data regarding income, poverty and income inequality. The 2021 Census data for income was collected and published in 2022, and the 2011 census data for income was collected and released in 2012. We filtered out the 2011 census, as it was irrelevant to our report, to help us effectively analyze our questions.

The variables in the first dataset containing the different income brackets, household mean incomes, and geographic information were removed; in the second dataset, all variables were removed with the exception of the ZIP codes, cumulative case rates, and cumulative death rates. Both datasets were then merged based on ZIP code, labeled as "ZIP", and subsequently filtered to only show data for years 2020-2021. The table has 25,404 observations. Each ZIP code includes multiple observations because the data was recorded weekly from March 1, 2020 and ended on January 1, 2022.

The variable "HouseMedian" displays the overall median household income, "FamilyMedian" displays the median household income for families, "MarriedMedian" displays the median of household income of married couples, "NonFamilyMedian" displays the median household income for nonfamilies, "Year" describes the Year that the data was collected, "CaseRateWeekly" displays the number of people per 100,000 that tested positive for COVID-19 for the week they were tested in, "DeathRateWeekly" displays the number of people per 100,000 that died due to COVID-19 for the specified week, "CaseRateCumulative" displays the number of people per 100,000 that tested positive for COVID-19, and "DeathRateCumulative" displays the number of people per 100,000 that died due to COVID-19.

ZIP	HouseMedian	FamilyMedian	MarriedMedian	NonfamilyMedian	Year	CaseRateWeekly	CaseRateCumulative	DeathRateWeekly	DeathRateCumulative
60601	123115.0	216810.0	211635.0	106488.0	2021.0	34.0	39448.0	0.0	102.2
60601	123115.0	216810.0	211635.0	106488.0	2021.0	27.0	39509.4	0.0	102.2
60601	123115.0	216810.0	211635.0	106488.0	2021.0	20.0	39529.8	0.0	102.2
60601	123115.0	216810.0	211635.0	106488.0	2021.0	34.0	38930.2	0.0	102.2
60601	123115.0	216810.0	211635.0	106488.0	2021.0	20.0	39686.5	0.0	102.2
...	...	...	...	...	...	...	...	...	...
60827	34878.0	42273.0	83585.0	28075.0	2020.0	4.0	4416.1	0.0	66.5
60827	34878.0	42273.0	83585.0	28075.0	2020.0	24.0	4377.6	0.0	66.5
60827	34878.0	42273.0	83585.0	28075.0	2020.0	0.0	4615.6	0.0	66.5
60827	34878.0	42273.0	83585.0	28075.0	2020.0	21.0	1060.3	0.0	45.5
60827	34878.0	42273.0	83585.0	28075.0	2020.0	14.0	4584.1	0.0	66.5



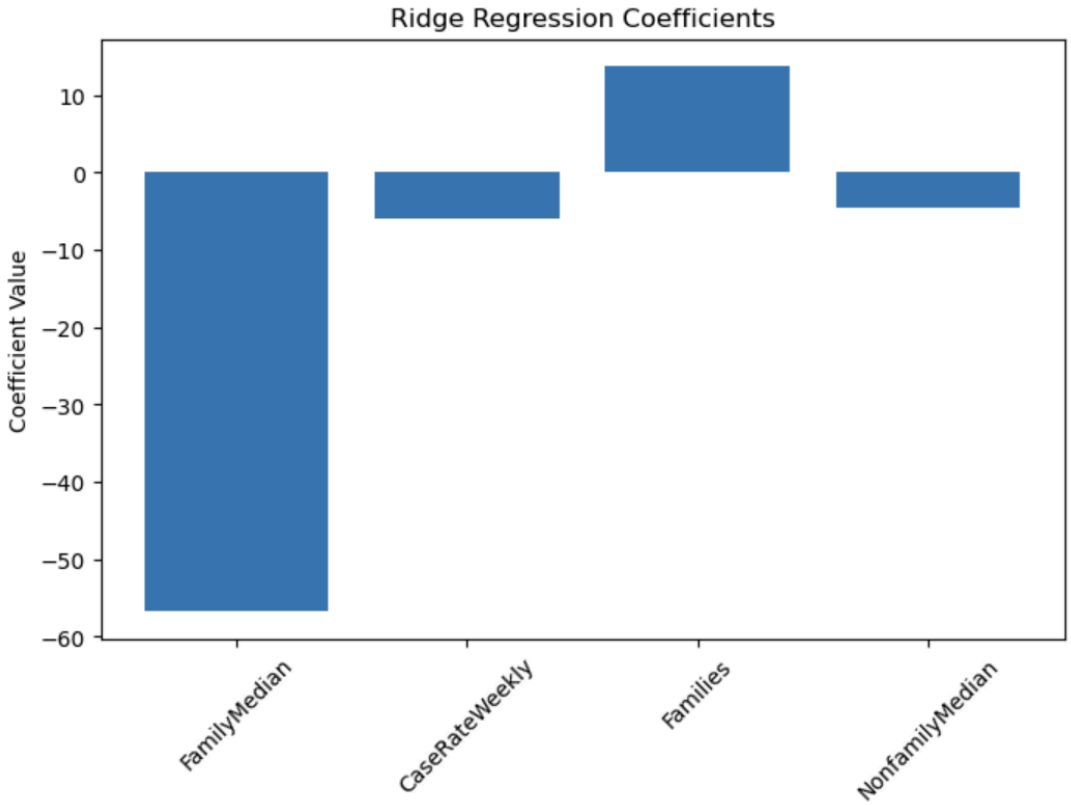
Results

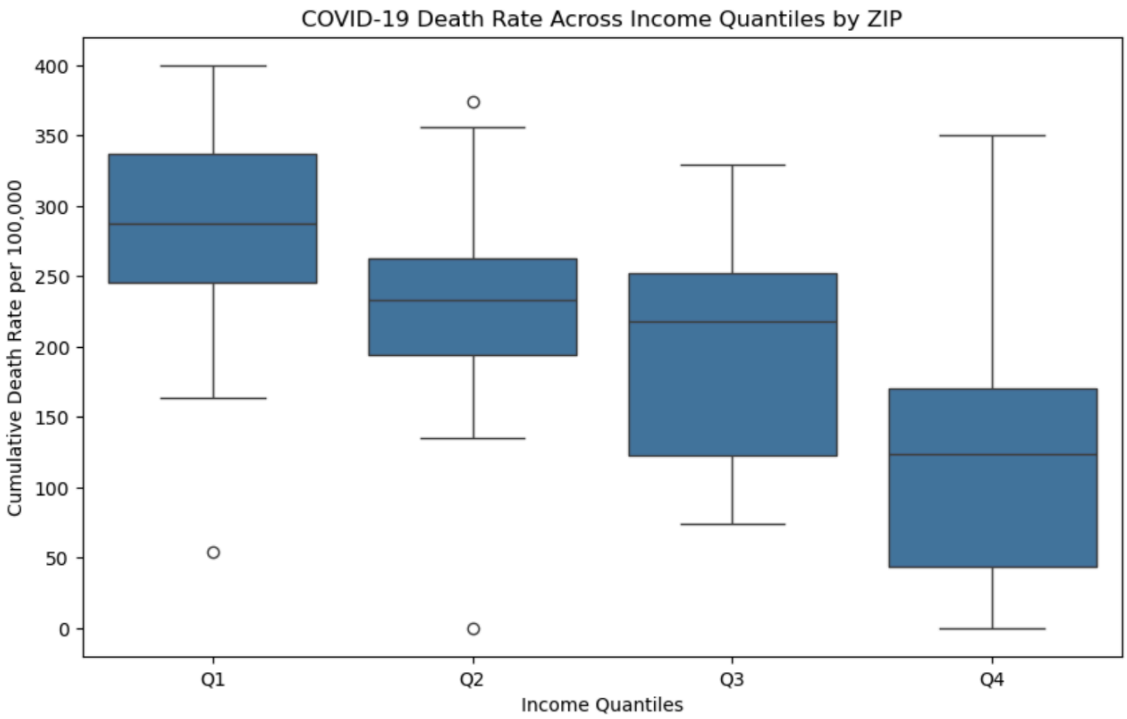
To investigate whether ZIP codes with lower median household incomes experienced higher COVID-19 death rates, we first prepared the dataset by merging COVID-19 mortality in Chicago ZIP codes with income statistics based on the Chicago ZIP codes. Then we filtered the combined data to the years 2020–2021. Because many demographic and income related variables were correlated, we assessed multicollinearity using the Variance Inflation Factor (VIF). The VIF results showed that FamilyMedian and NonfamilyMedian had moderately high VIF values (around 5), while Families and CaseRateWeekly were close to 1. This means there is some overlap among the income related variables, but not enough to make them unusable. Therefore it showed that although some income variables were related, they were not collinear enough to prevent regression modeling. This step ensured that the following analysis would generate stable, interpretable coefficient estimates.

After confirming the predictors were acceptable to use, we built a Ridge Regression model to examine how each variable contributed to predicting cumulative death rates. The Ridge Regression model was chosen because it handles moderate multicollinearity better than standard linear regression by shrinking the coefficients toward zero. The model produced a mix of positive and negative coefficients: Families had a positive coefficient, while FamilyMedian, CaseRateWeekly, and NonfamilyMedian had negative coefficients. From this model, ZIP codes with more family households predicted higher death rates because the

coefficient value for families is positive, so we know that as the value for families goes up, the death rate also increases. In contrast, ZIP codes with lower median incomes tended to predict higher death rates since negative coefficients indicate that death rates increase as income decreases. To complement the regression results, we created a box plot visualization by separating ZIP codes into income quartiles using q-cut and plotting the cumulative death rates across these groups. The box plot revealed a clear pattern that ZIP codes in the lowest income quartile (Q1) had noticeably higher death rates than those in the highest quartile (Q4), as evidenced by a consistent drop in death rates as income increased. This visualization clearly supported the model results. Moreover, the box plot for Q1 has a median death rate of about 290 per 100,000, with the distribution skewed towards higher values and an interquartile range of approximately 250-335. On the other hand, the Q4 (highest income) quartile had a median death rate of about 120 per 100,000, fewer than half that of Q1. The Q2 and Q3 quartiles showed median values at approximately 230 and 210 respectively, demonstrating a clear linear decrease in death rates as income increased.

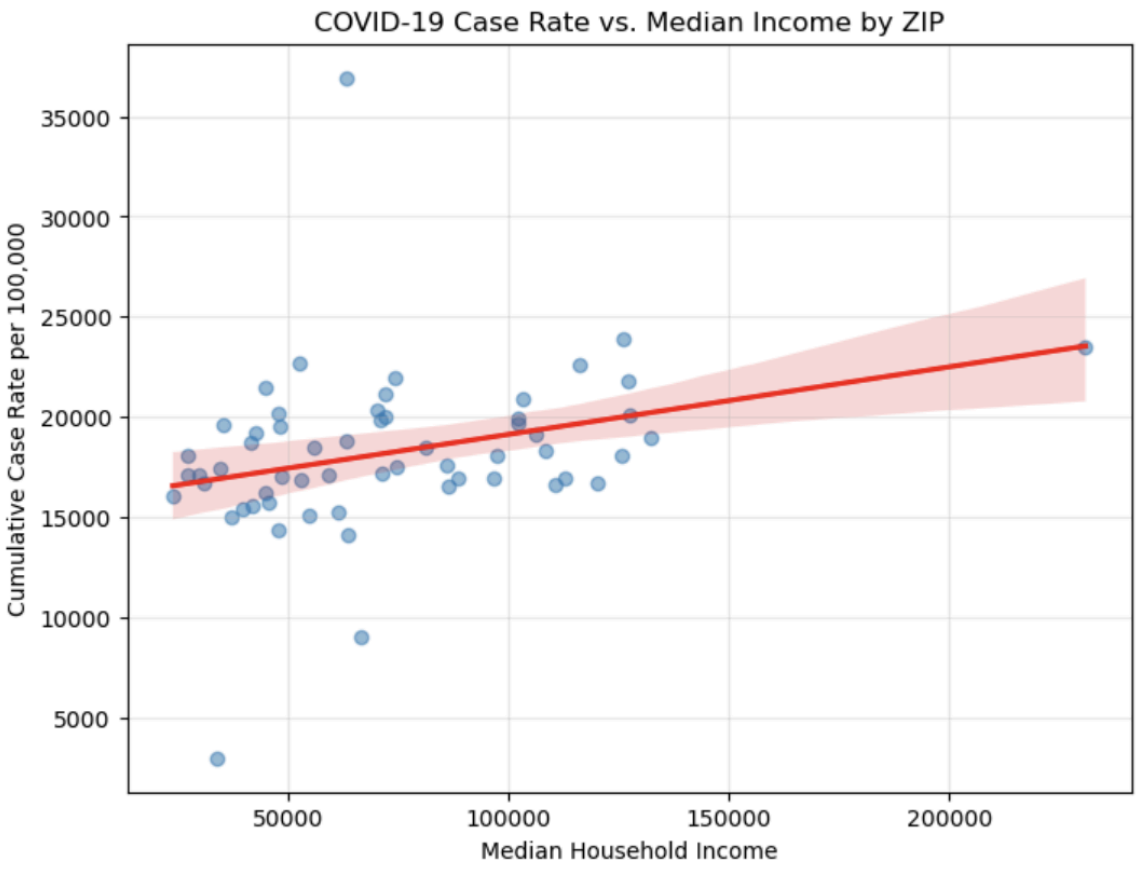
Based on the Ridge Regression model and the box plot, there is a pattern that ZIP codes with low incomes experienced much higher COVID-19 death rates. Although the regression coefficients were influenced by overlapping variables, they still supported the visual evidence of the box plot and hence strongly support our hypothesis that lower-income ZIP codes in Chicago from 2020-2021 bore a disproportionate mortality burden. This difference in median death rates suggests that socioeconomic factors played a critical role in COVID-19 outcomes.

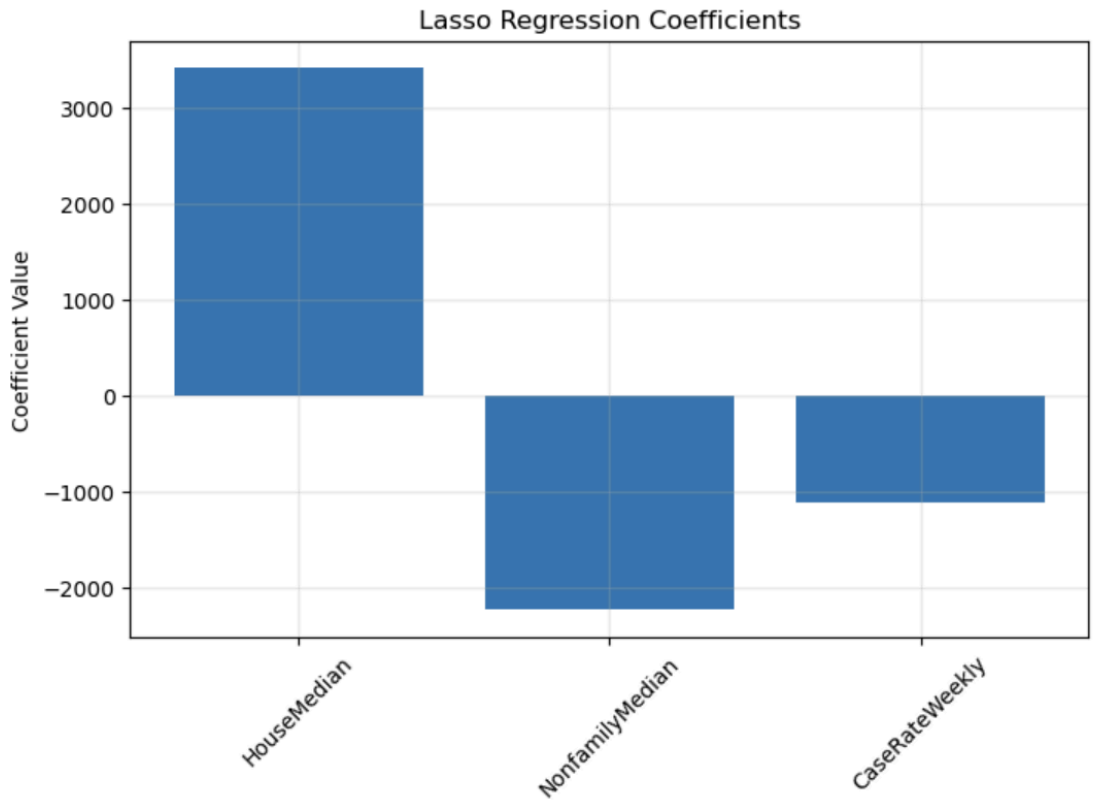




To explore whether ZIP codes with lower median household incomes experienced higher cumulative COVID-19 case rates, two approaches were conducted. The first method was a linear regression plot for visual examination of the relationship between median household income and cumulative case rates across all ZIP codes. This scatter plot included a fitted trend line (in red) with a confidence interval (the shaded red region) to show the overall direction and strength of the association. The regression plot displayed a positive correlation between median household income and cumulative COVID-19 case rates. Lower-income ZIP codes (around 20,000 - 50,000 median income) typically had case rates between 15,000-20,000 per 100,000 residents, while higher-income areas (above 100,000 median income) showed case rates from about 17,000 to 23,500 per 100,000. At higher income levels, the shaded confidence interval grew wider, which indicates more variability in case rates as the median income grew. There were a few outliers, including one ZIP code with an extremely high case rate around 37,000 per 100,000 and another with a very low rate around 3,000 per 100,000. The second approach was Lasso regression, which is similar to standard linear regression but includes a penalty that can shrink less important predictor coefficients to zero. This allows us to identify which variables are important for predicting case rates. The Lasso model included three predictors: median household income, median income for non-family households, and weekly case rates. The results show that "HouseMedian" had the largest coefficient, about 3,400, meaning it was the strongest predictor. This also means that higher income strongly predicted higher cumulative case rates. On the other hand, "NonfamilyMedian" showed a negative coefficient, about -2,200, suggesting that areas with higher non-family household incomes had somewhat lower case rates. This could be due to the fact that nonfamily households have different structures or demographic patterns than family households. CaseRateWeekly had a negative coefficient of roughly -1,050, which most likely reflects how the pandemic unfolded at different times in different neighborhoods.

Through analyzing the visualizations, lower-income ZIP codes did not have higher cumulative COVID-19 case rates. Instead, the data showed the opposite pattern that wealthier ZIP codes consistently reported higher case rates per 100,000 residents. Both the visual regression plot and the Lasso model seems to conclude that as income increased, reported case rates increased. These finding seem to contradict popular beliefs, especially when combined with our first finding that lower-income areas had much higher death rates. The most likely explanation is that this pattern reflects differences in testing access rather than true infection rates. Wealthier residents in Chicago were more likely to have better access to COVID-19 testing throughout the pandemic, were more likely to visit doctors when symptomatic, had employers or insurance that provided regular testing, and are around better healthcare infrastructure. From the seemingly unusual result of the visualizations, some possiblities for this contradictory conclusion can be due to barriers that prevented testing COVID-19 in lower-income neighborhoods. These results may be attributed to the lack of health insurance, inability to miss work for testing appointments, fewer nearby testing locations, and medical costs. This means the true infection rate in lower income ZIP codes could possibly be much higher than what the data shows.





Conclusion

In this report, our goal was to provide an in-depth analysis and answer the questions "Do ZIP codes with lower median household incomes have higher COVID-19 death/mortality rates?" and "Do ZIP codes with lower median household incomes have higher cumulative COVID-19 case rates?" to address possible disparities in how the COVID-19 pandemic affected people's lives across income ranges. Through this analysis, we discovered through Ridge Regression that there is a clear linear decrease in death rates as income increased, indicating that our original hypothesis of lower median income households having higher death rates was true. In contrast, we also discovered through a linear regression plot and Lasso regression that wealthier ZIP codes consistently reported higher case rates, indicating that our original hypothesis that ZIP codes with lower median household incomes would have higher cumulative COVID-19 case rates was false.

These findings are important because they highlight significant differences between income levels in Chicago, and possibly the entire United States, seeing as Chicago has a large population size. Though this is not an adequate, official generalization of the United States, this investigation did having a large sample size could unofficially indicate a trend in the United States as a whole. Seeing as the death rates in lower-income levels were higher, one could investigate further into the causes behind this relationship, and possibly prevent some deaths and illnesses in people in lower-income levels during pandemics or general sickness waves. If a cause of this negative correlation between income level and death rate is a lack of access to healthcare, then this is a preventive problem that people should care about to

ensure equal treatment, and those with the resources, like politicians, activists, free clinic workers, etc. should advocate for more equal treatment of people when it comes to COVID-19, despite what income level they may be in. This activism could be the key to preventing future deaths and helping people of lower-income levels get better. Additionally, as mentioned before, it is a possibility that the positive correlation between higher-income levels and COVID-19 case rates is a result of better access to testing centers and tests in general, rather than a higher rate of infection in higher-income levels. This would indicate that many people in lower-income levels are unable to get tested, and therefore unable to get the proper treatment they deserve for their symptoms. Of course, this is an assumption, but the risk of this being a possibility shows a clear need for these findings to be looked into further.

In the future, one could continue to look into these results and this work in total by running these analyses again with different models or methods and initializing similar tests with different datasets. In our investigation, we did not have demographic or quantitative data available regarding healthcare access or socioeconomic factors; having this data would have refined our findings and our interpretations, especially considering our belief that a cause behind both correlations studied is a lack of access to healthcare. When analyzing this data, clustering by these factors, or other factors that we actually did have in our data like the different kinds of medians, could improve the results. Additionally, to expand on these multiple medians, they could have been treated as different variables and therefore we could have used Multiple Linear Regression as a model. Though this was not part of our question, it would have been interesting to analyze if these conditions had improved between 2020 and 2021 using a training split versus a test split. Looking at the differences between these two splits could have shown any improvement or decrease in conditions, and therefore researchers could have spotted what solutions worked and didn't work when it came to addressing the problem these findings imply regarding socioeconomic status. Furthermore, to help generalize these findings across the United States, or at least in different areas, these same tests could be repeated on different populations by their zip codes, as in various cities or states. All in all, while our findings and methods could be refined, our data analysis revealed strong findings that could help us and other researchers further understand the impacts of the COVID-19 pandemic across different income levels and assess how to solve any problems caused by these impacts.